Literature Status: Near Unconditionality versus Truncation Quasi-Greediness

Run `fa_banach_001` — lane 3

21 July 2026

Status: literature already answered. Question 5.1 of arXiv:2209.03445 asks whether a nearly unconditional basis can fail to be truncation quasi-greedy. The same authors explicitly answer this in the affirmative in the separate later paper arXiv:2305.12253: their Theorems 1.5 and 1.6 produce such a Schauder basis in a Banach space.

1 Original question

F. Albiac, J. L. Ansorena, and M. Berasategui, *Elton's near unconditionality of bases as a threshold-free form of greediness*, arXiv:2209.03445, later published in *Journal of Functional Analysis* **285** (2023), 110060, poses the following on arXiv PDF page 19 [1].

Source Question 5.1. *Is there a nearly unconditional basis that is not truncation quasi-greedy?*

The source explains that near unconditionality is equivalent to quasi-greediness for largest coefficients (QGLC), whereas truncation quasi-greediness was known to imply QGLC. The question asks whether the converse can fail.

2 Separate later answer

The same three authors subsequently wrote *Linear versus nonlinear forms of partial unconditionality of bases*, arXiv:2305.12253, published in *Journal of Functional Analysis* **287** (2024), 110594 [2]. On arXiv PDF page 5 they write, immediately before Theorem 1.6, that they solve the question in the negative—that is, not every QGLC basis is truncation quasi-greedy.

Their Theorem 1.5 states that, for a basis of a quasi-Banach space, the following are equivalent:

$$\text{bounded-oscillation unconditionality} \iff \text{truncation quasi-greediness.}$$

Their Theorem 1.6 then states:

Later answer. *There is a Schauder basis of a Banach space that is nearly unconditional but is not bounded-oscillation unconditional. Moreover, for every $1 \le p < \infty$ the example may be chosen with signed fundamental function comparable to $m^{1/p}$.*

Combining the two theorems, this basis is nearly unconditional but not truncation quasi-greedy. It is therefore exactly the counterexample requested by Question 5.1. This implication is also the authors' stated purpose in the paragraph introducing Theorem 1.6, so the identification is explicit rather than an agent-inferred consequence.

3 Scope and search status

The later construction completely answers the existence question in Question 5.1, and does so already in a Banach space. It does not by itself settle Question 5.2 of the source concerning equivalent renormings and uniform QGLC constants.

A bounded search through 21 July 2026 checked the run registry, local arXiv corpus, exact question wording, QGLC/truncation terminology, and later work by the source authors. The decisive match is arXiv:2305.12253 and its published JFA version. No new mathematical result is claimed in this packet.

References

[1] F. Albiac, J. L. Ansorena, and M. Berasategui, *Elton's near unconditionality of bases as a threshold-free form of greediness*, J. Funct. Anal. **285** (2023), 110060; arXiv:2209.03445.

[2] F. Albiac, J. L. Ansorena, and M. Berasategui, *Linear versus nonlinear forms of partial unconditionality of bases*, J. Funct. Anal. **287** (2024), 110594; arXiv:2305.12253.